AutoDrive

Autonomous Driving Research Prototype

Computer vision, lane segmentation and steering prediction



AutoDrive Research Team

github.com/HA2345567/Autodrive

Project team and academic guidance

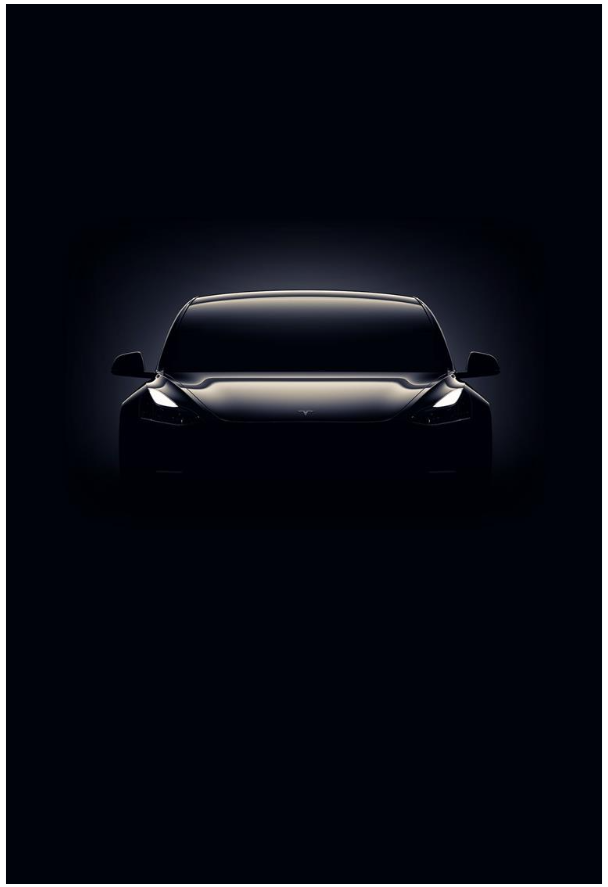
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Presentation outline



01 Problem and project objectives

02 Architecture, data and training

03 Detection, lane and steering results

04 Limitations, applications and next steps

Introduction

One front-camera frame supports three separately evaluated tasks

01 Object detection identifies road users and traffic infrastructure.

02 Lane segmentation estimates the pixels belonging to lane markings.

03 Steering regression estimates a continuous steering angle.

04 A visual display combines the outputs for inspection.

Project objectives

01 Train a compact detector for ten road-relevant COCO categories.

02 Convert TuSimple lane points into polygons for YOLO segmentation.

03 Audit the steering checkpoint and evaluate a new frozen-CNN readout.

04 Document each module’s accuracy, timing and evaluation limits.

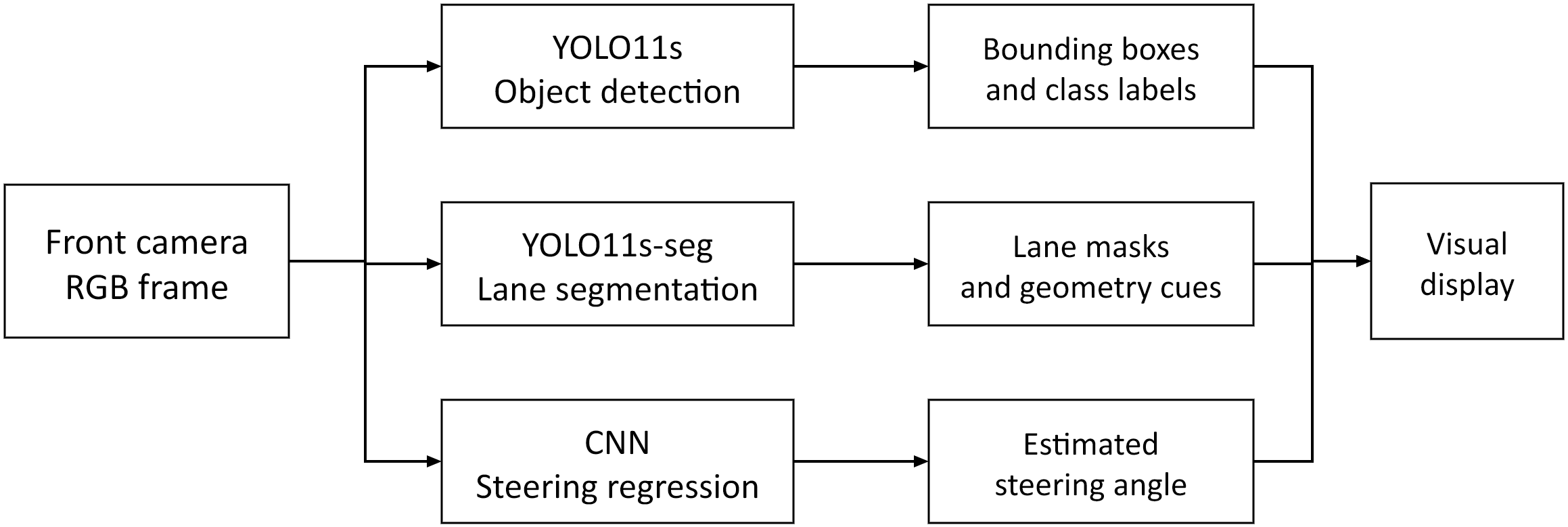
Literature review

Work	Contribution	Relevance to AutoDrive
DAVE-2, Bojarski et al. (2016)	Camera images to steering angle	Reference architecture for steering regression
SCNN and LaneNet (2018)	Lane structure and instance segmentation	Context for learning lane markings
Ultralytics YOLO11 (2024)	Object detection and instance segmentation	Implemented perception models
TuSimple benchmark (2017)	Lane annotations and evaluation protocol	Lane dataset and metric caveats

Contribution: integration, annotation conversion and component evaluation

Integrated AutoDrive architecture

Parallel branches process the same camera frame



Research prototype. No validated closed-loop vehicle control.

Datasets and preprocessing

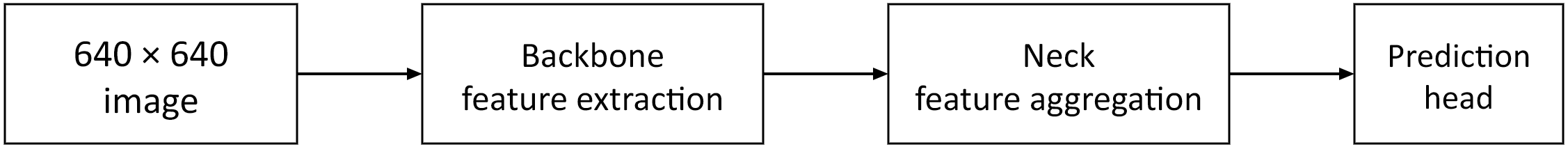
Module	Training data	Evaluation data	Preprocessing
Detection	8,080 COCO-10 images	975 validation + 525 test	640 × 640 input, ten classes
Lane segmentation	2,900 TuSimple frames	726 validation frames	Lane points converted to narrow polygons
Steering	36,324 ordered training rows	9,082 validation rows	Bottom 150-row crop, 200 × 66 resize

Lane annotations: 3,626 valid frames and 2,732 validation lane instances

The existing steering split is not an independent test drive.

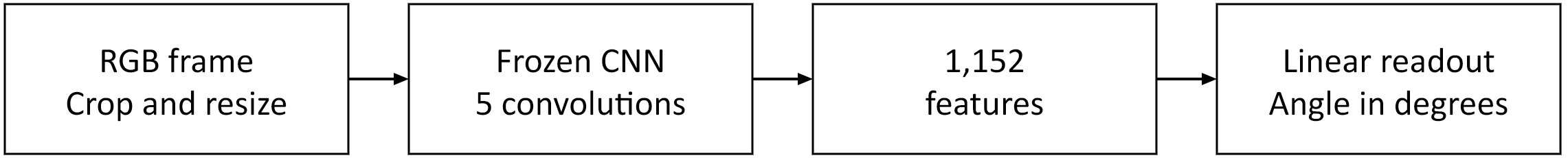
Perception models

Model	Task	Parameters	Fused compute
YOLO11s	10-class object detection	9,416,670	21.4 GFLOPs
YOLO11s-seg	1-class lane masks	10,067,203	32.9 GFLOPs



Detection: boxes and classes. Segmentation: boxes, masks and confidence.

Steering model and new readout



01 The saved CNN contains 1,595,511 parameters, including a 1,164-unit dense layer.

02 The new experiment freezes the backbone and fits a Huber-loss readout.

03 The canonical 252,219-parameter DAVE-2 remains a reference configuration.

Training setup and requirements

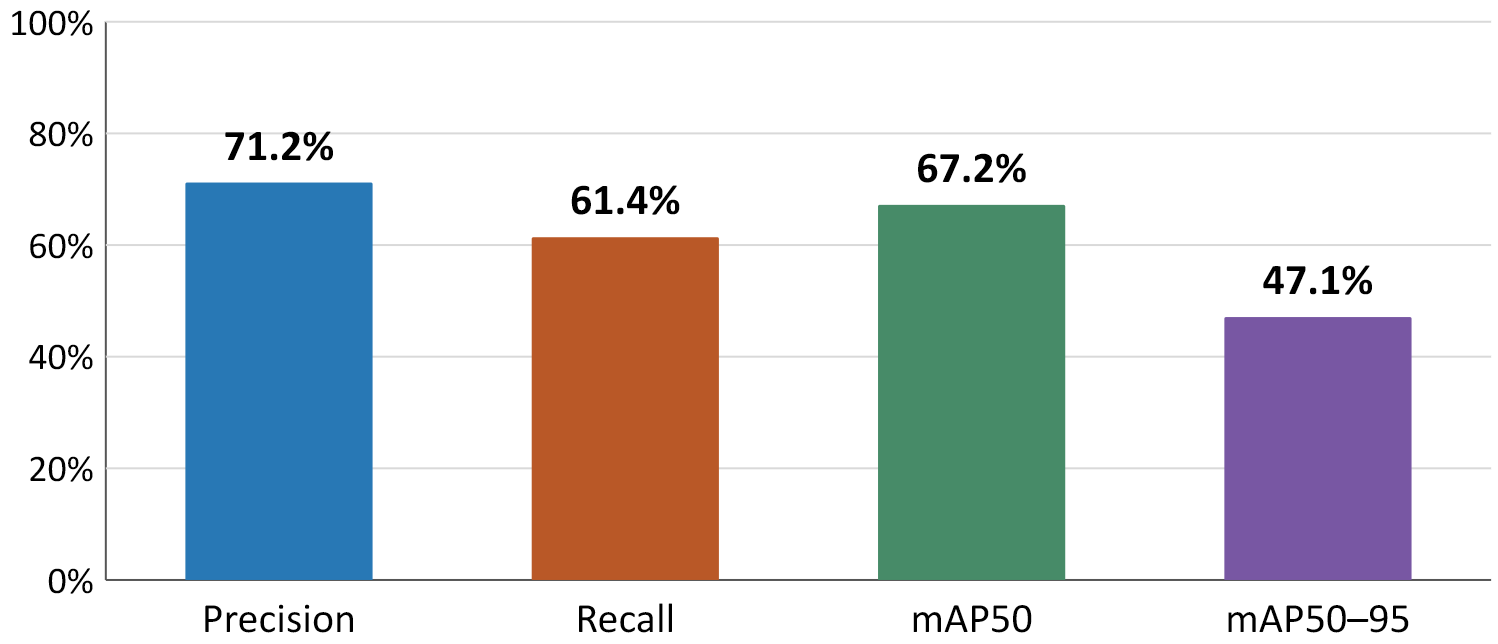
Run	Epochs	Batch	Input	Hardware / time
Detection	50	16	640 × 640	Colab Tesla T4 / 2.14 h
Lane segmentation	60	16	640 × 640	Colab Tesla T4 / 2.31 h
Original steering setup	30	100	66 × 200	Historical training time not recorded

Software: Python, PyTorch, Ultralytics, TensorFlow, OpenCV and NumPy

New steering readout: local Intel Pentium 4415U CPU

Object detection results

Held-out test: 525 images and 3,116 annotated instances



71.2%

precision

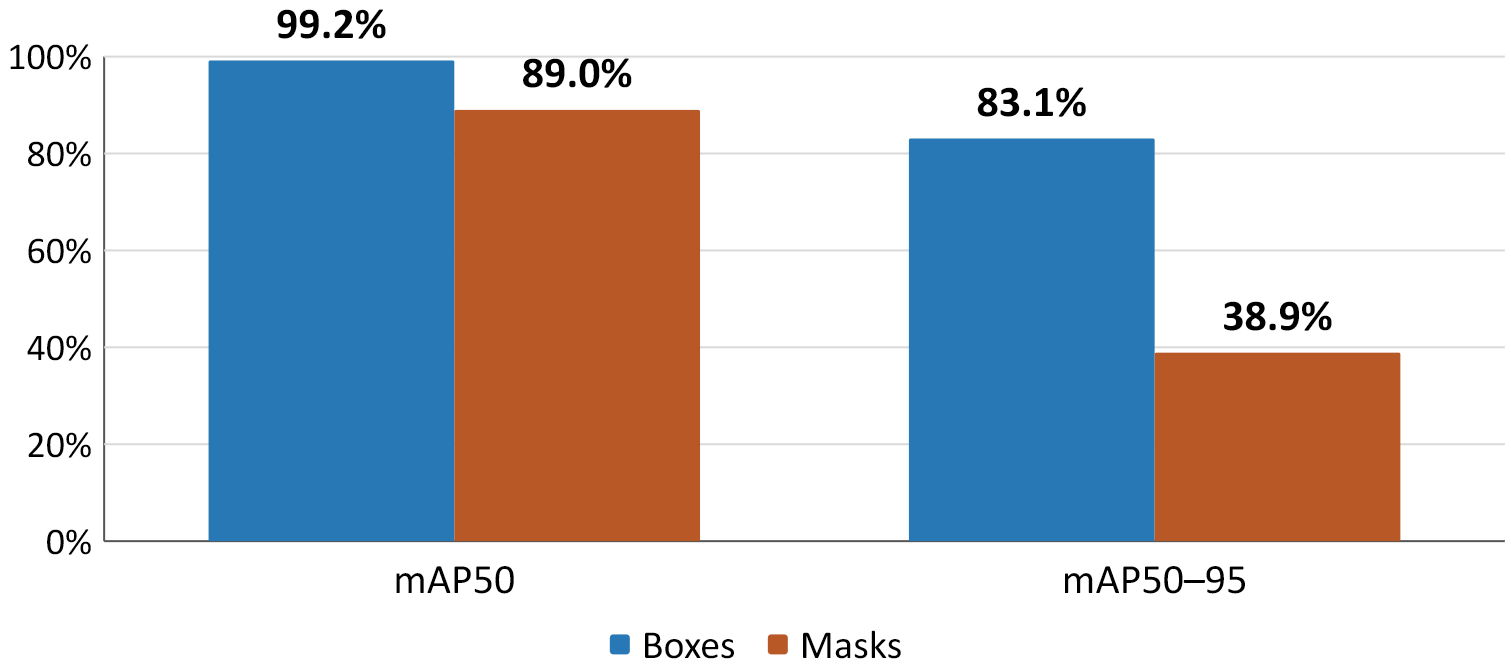
61.4%

recall

mAP is a detection metric. It is not classification accuracy.

Lane segmentation results

Validation: 726 frames and 2,732 annotated lane instances



99.2%

box mAP50

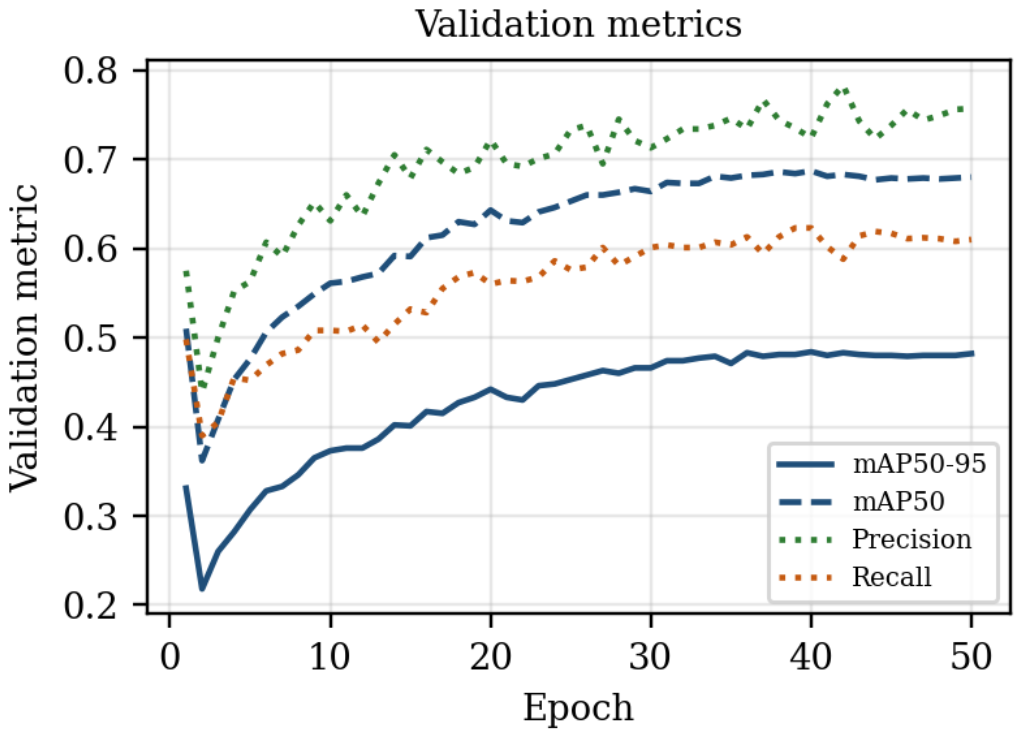
38.9%

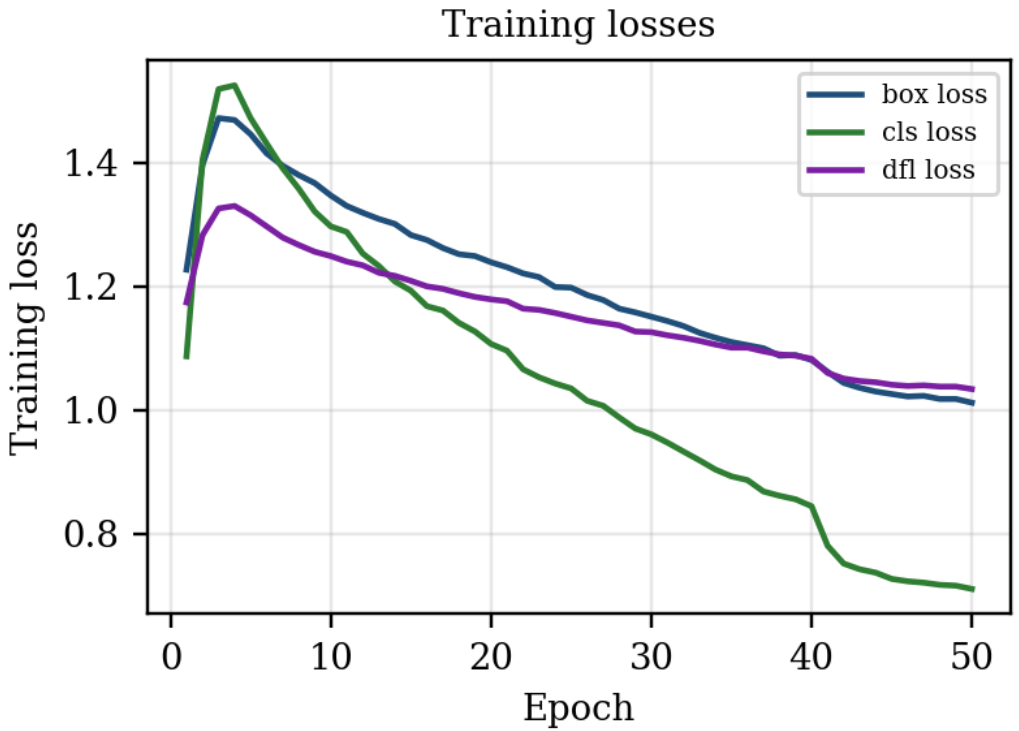
mask mAP50–95

Thin lane masks make IoU sensitive to small offsets. These are not official TuSimple leaderboard scores.

Detection training progress

50 recorded epochs on the perception training run

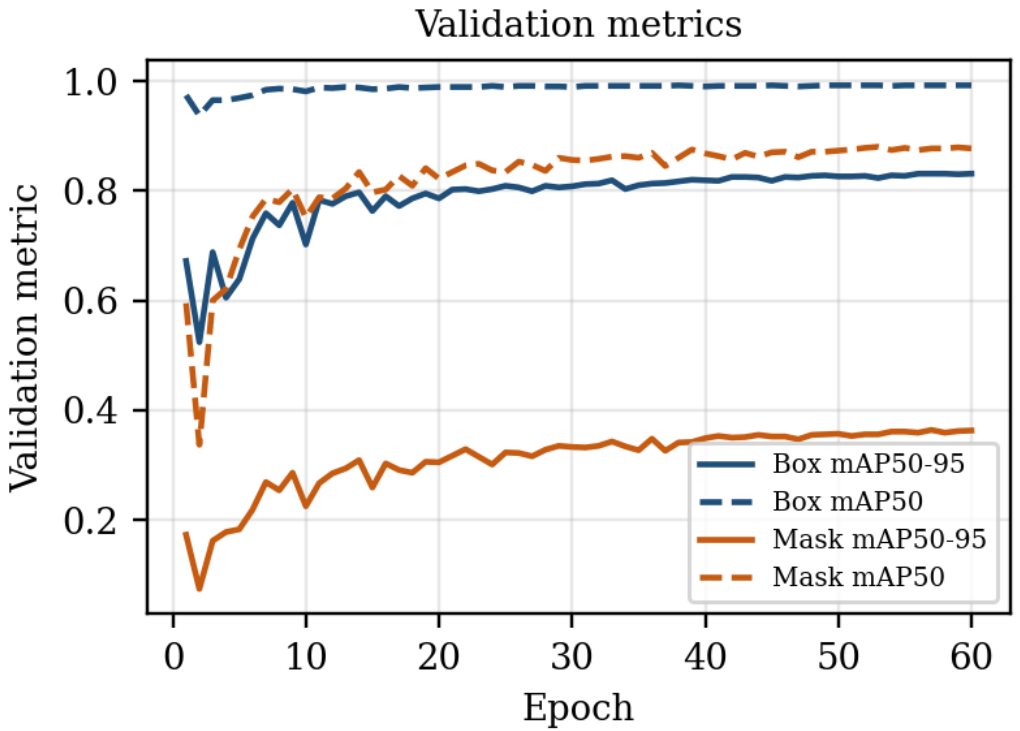


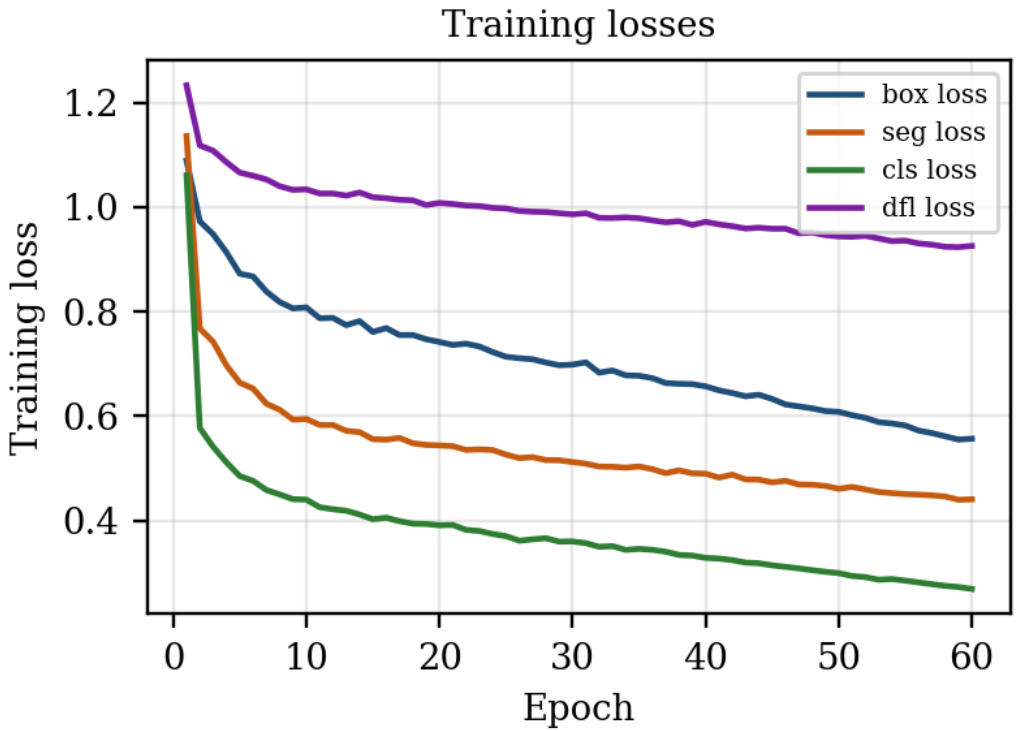


Validation mAP50–95: 0.331 at epoch 1, 0.482 at epoch 50

Lane training progress

60 recorded epochs on the perception training run

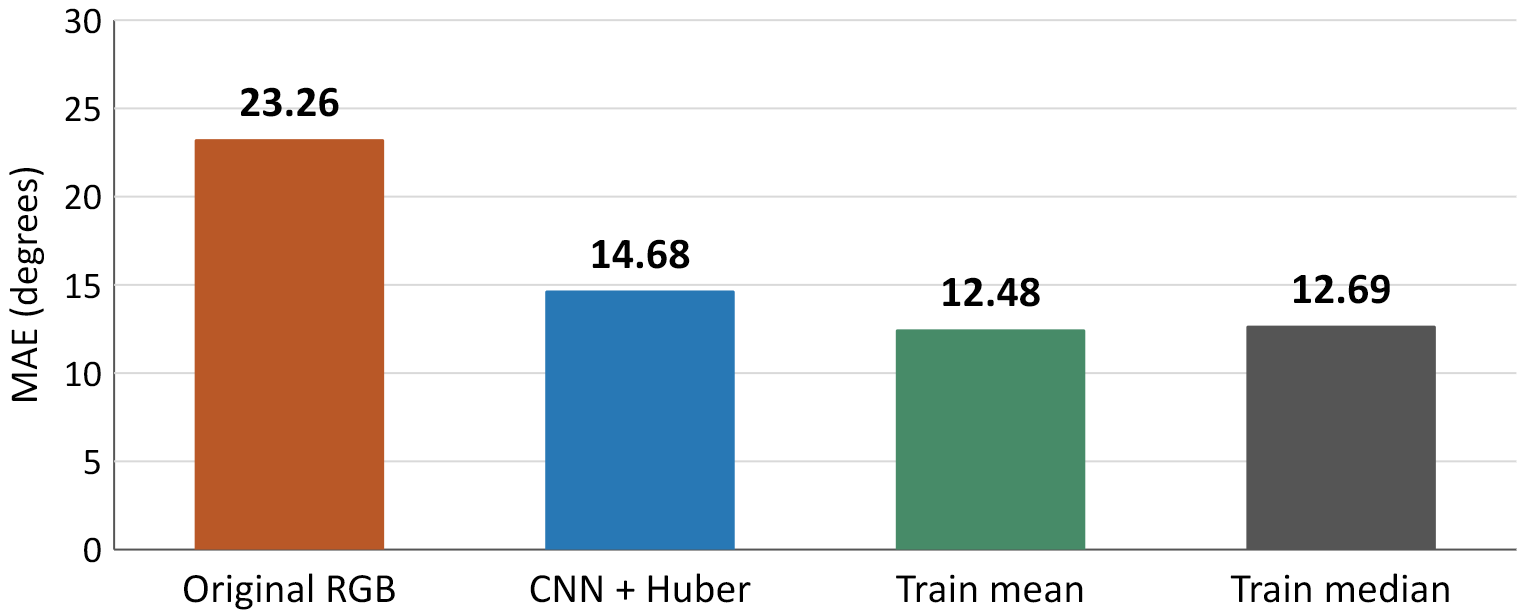




Mask mAP50–95: 0.173 at epoch 1, 0.363 at epoch 60

Steering results and baselines

Same 9,082 validation frames for all four models



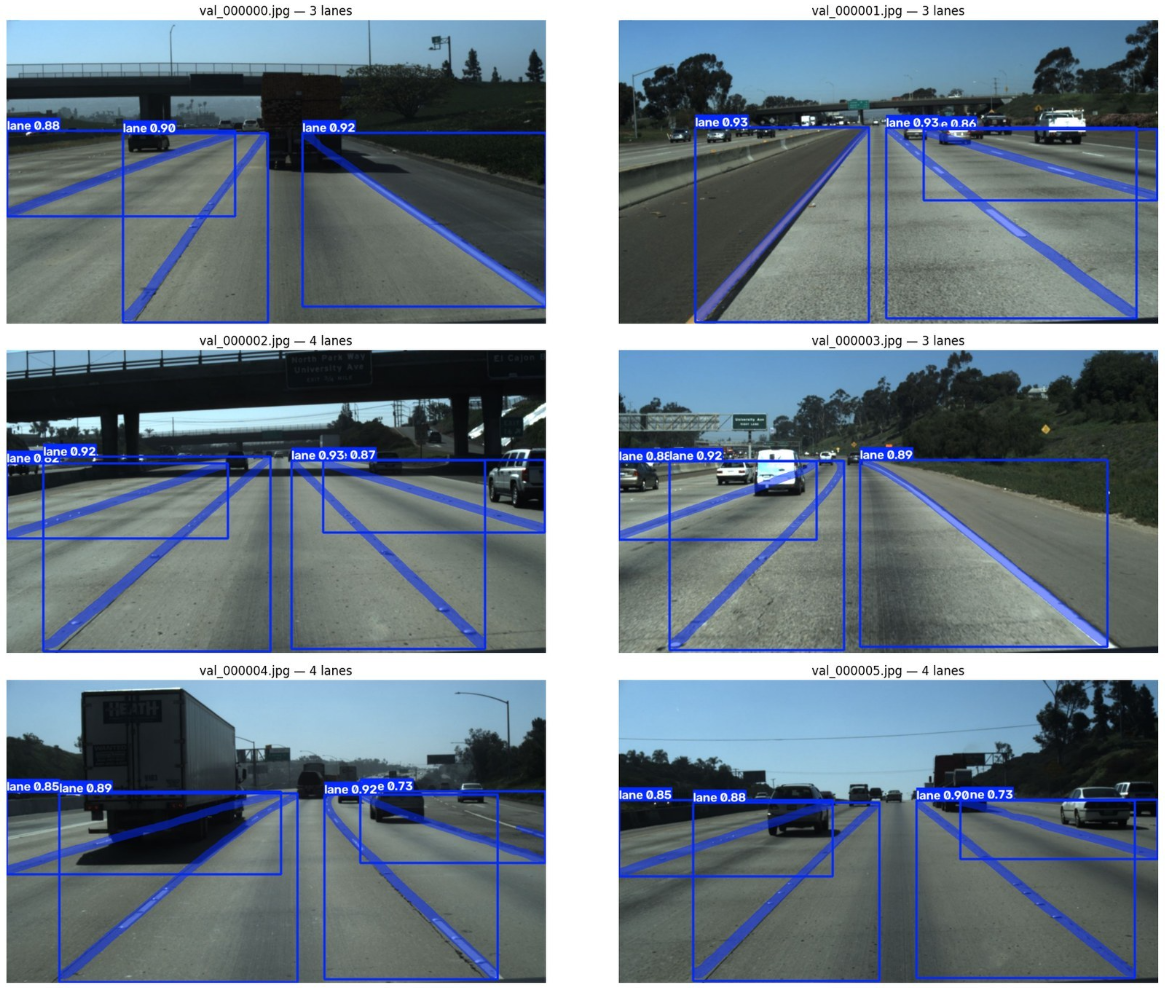
36.9%

lower MAE than
the original RGB CNN

14.6822° MAE
R^2 = –0.1336

Constant baselines remain stronger. Reliable image-conditioned steering is not established.

Lane prediction examples



What the image shows

Predicted lane boxes and masks on six validation frames.

Examples include shadows, underpasses and a truck occlusion.

Qualitative evidence complements the numerical metrics.

Object prediction examples



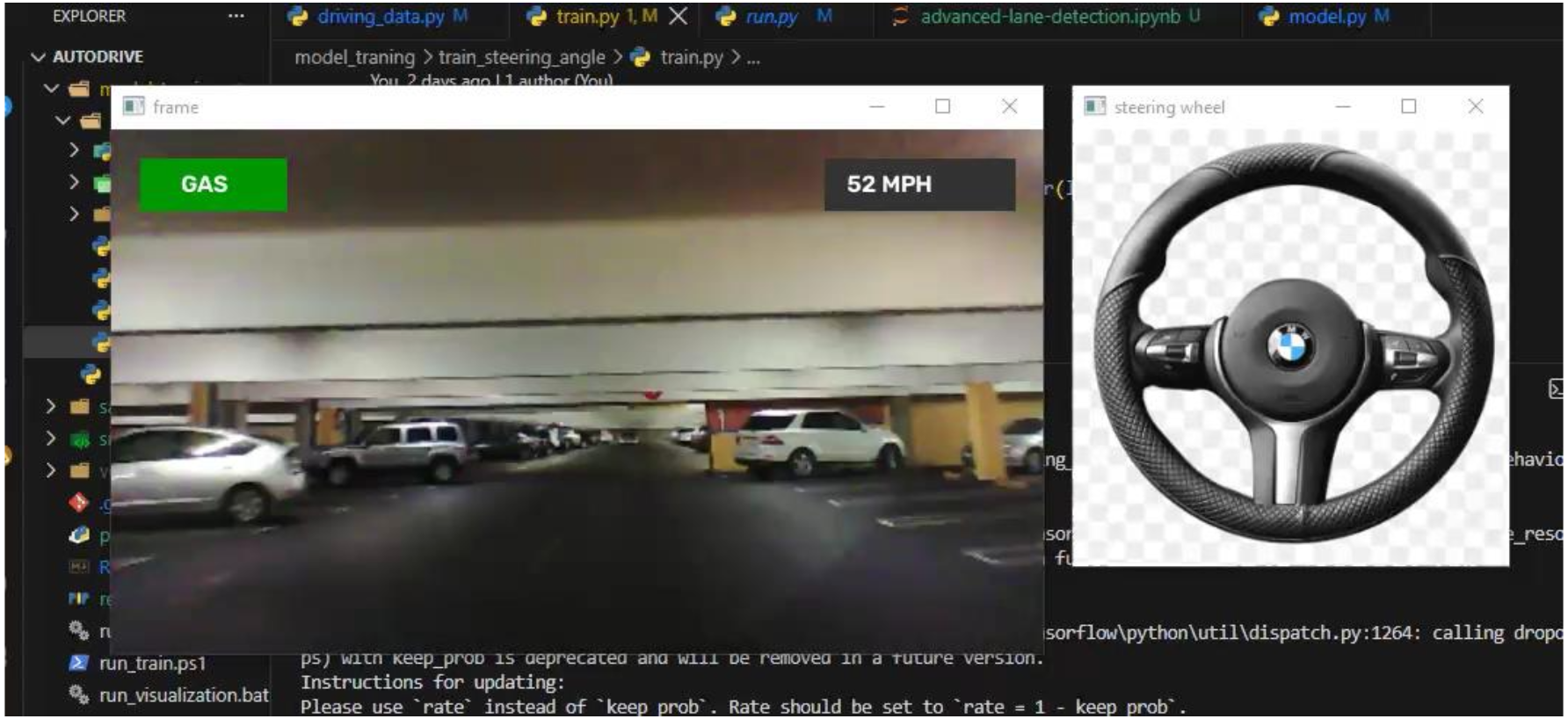
What the image shows

Bounding boxes, class names and confidence scores.

The detector finds large objects well, but small objects and recall remain challenges.

Test images remain separate from training.

Prototype visualization



Display demonstration only. Speed and pedal indicators do not establish vehicle control.

Latency and model footprint

Separate framework timings on a Tesla T4

Timing mode	Detection	Lane	Combined
Batch-16 evaluation	11.1 ms	11.8 ms	22.9 ms
Single-image prediction	17.0 ms	14.3 ms	31.3 ms
Saved weights	19.2 MB	20.5 MB	39.7 MB

32–44 FPS

Projected throughput for the two YOLO modules only

Full pipeline FPS was not measured. Capture, steering, display and control I/O are excluded.

Challenges, applications and next steps

Current limitations

01 Steering trails simple constant baselines.

02 Lane scores use synthetic masks.

03 Independent-drive and closed-loop tests remain outstanding.

Research applications

Component benchmarking
Lane annotation experiments
Perception and visualization teaching

Next steps

Official lane metrics, new driving data and controlled simulator evaluation

Conclusion and proof of work

01 AutoDrive integrates detection, lane segmentation and steering prediction.

02 Perception results are useful, while steering reliability remains unresolved.

03 The implementation and experiment artifacts support reproducibility.

github.com/HA2345567/Autodrive

Thank you. Questions are welcome.

References

[1] Bojarski et al. End to End Learning for Self-Driving Cars. arXiv:1604.07316, 2016.

[2] Jocher and Qiu. Ultralytics YOLO11. Ultralytics repository, 2024.

[3] TuSimple. TuSimple Lane Detection Benchmark. Dataset and evaluation code, 2017.

[4] Lin et al. Microsoft COCO: Common Objects in Context. ECCV, 2014.

[5] Neven et al. Towards End-to-End Lane Detection: An Instance Segmentation Approach. IV, 2018.

[6] AutoDrive Research Team. AutoDrive research paper and implementation repository.

Appendix A COCO-10 dataset counts

Instances per class. Image split: 8,080 train / 975 validation / 525 test

Class	Train instances	Train share	Val instances	Test instances
person	24,661	43.8%	2,728	1,567
bicycle	2,887	5.1%	210	104
car	12,443	22.1%	1,276	642
motorcycle	3,054	5.4%	220	147
bus	2,188	3.9%	169	114
truck	2,914	5.2%	267	147
traffic light	4,768	8.5%	403	231
stop sign	1,118	2.0%	53	22
dog	1,221	2.2%	146	72
cat	1,013	1.8%	132	70
Total	56,267	100%	5,604	3,116

Appendix B Per-class detection results

Held-out test results. Validation AP50–95 is shown for comparison.

Class	Test P	Test R	Test AP50	Test AP50-95	Val AP50-95	Test inst.
person	0.812	0.613	0.717	0.457	0.469	1,567
bicycle	0.686	0.399	0.498	0.270	0.311	104
car	0.763	0.607	0.680	0.456	0.406	642
motorcycle	0.802	0.633	0.771	0.472	0.422	147
bus	0.842	0.711	0.791	0.662	0.669	114
truck	0.561	0.463	0.521	0.331	0.347	147
traffic light	0.636	0.476	0.494	0.265	0.271	231
stop sign	0.488	0.607	0.619	0.556	0.736	22
dog	0.651	0.806	0.771	0.596	0.558	72
cat	0.882	0.829	0.861	0.645	0.667	70

Appendix C Full perception metrics

Detection uses a held-out test. Lane results use validation only.

Split	Images	Instances	Precision	Recall	mAP50	mAP50-95	mAP75
Epoch-1 reference (val)	975	5,604	0.575	0.498	0.510	0.331	–
Validation	975	5,604	0.724	0.621	0.688	0.486	0.518
Test (held out)	525	3,116	0.712	0.614	0.672	0.471	0.511

Lane segmentation

Metric	Epoch 1	Training end	Re-evaluation
Box P	0.955	0.976	0.976
Box R	0.953	0.974	0.974
Box mAP50	0.974	0.992	0.992
Box mAP50-95	0.672	0.831	0.831
Mask P	0.744	0.918	0.925
Mask R	0.735	0.916	0.923
Mask mAP50	0.595	0.877	0.890
Mask mAP50-95	0.173	0.363	0.389

Appendix D Complete steering metrics

All four models below use the same 9,082-frame validation partition.

Model	MAE °	RMSE °	Bias °	R^2	Pearson r	Within ±5°
Original CNN (RGB)	23.2570	33.6313	+21.4106	-0.8067	0.0508	9.82%
Frozen CNN + Huber	14.6822	26.6395	+3.1230	-0.1336	0.1682	31.35%
Training-mean constant	12.4793	25.0502	+1.2196	-0.0024	Undefined	42.57%
Training-median constant	12.6900	25.0212	-0.1859	-0.0001	Undefined	41.62%

Legacy BGR audit: 505 frames sampled every 18 rows

MAE 22.9616° RMSE 33.2350° Bias +20.9977°
R^2 −0.8769 Pearson r −0.0404 Within ±5° 9.11%

Appendix E Recorded experimental comparisons

Saved notebook outputs. These are not controlled ablation results.

Comparison	Metric	Reference	Observed	Change
Detection: epoch 1 to 50	Validation mAP50-95	0.331	0.482	+0.151
Lane: epoch 1 to 60	Box mAP50-95	0.672	0.831	+0.159
Lane: epoch 1 to 60	Mask mAP50-95	0.173	0.363	+0.190
Lane: checkpoint re-evaluation	Mask mAP50-95	0.363	0.389	+0.026
Detection: validation to test	mAP50-95	0.4855	0.4711	-0.0144
Detection: validation to test	mAP50	0.6883	0.6723	-0.0160

Deltas use the precision printed in the source notebooks.

Appendix F Training hyperparameters

Original module training configuration from the recorded runs

Parameter	Detection (YOLO11s)	Lane (YOLO11s-seg)	Steering (DAVE-2)
Initial weights	yolo11s.pt (COCO)	yolo11s-seg.pt	Random initialization
Epochs / patience	50 / 15	60 / 20	30 / none
Batch size	16	16	100
Input size	640×640	640×640	66×200
Optimiser	auto	auto	Adam
Loss	CIoU + BCE + DFL	CIoU + BCE + DFL + mask	MSE
LR schedule	cosine, 3 warm-up ep.	linear, 3 warm-up ep.	Fixed 1e-4
Weight decay	5×10−4	5×10−4	L2 0.001
Mosaic	1.0 (off last 10 ep.)	0.0	-
Flip / scale / translate	0.5 / 0.5 / 0.1	0.5 / 0.4 / 0.1	None
HSV (h, s, v)	0.015, 0.7, 0.4	0.015, 0.7, 0.4	-
Rotation / shear / perspective	default	0 / 0 / 0	-
mask_ratio	–	2	-
Dropout	0	0	Keep probability 0.5
Seed / deterministic	0 / yes	0 / yes	Not set
Mixed precision	yes	yes	No
Training time (T4)	2.14 h	2.31 h	Not recorded

Appendix G Steering experiment protocol

Development selection stays within the original training partition.

Item	Recorded value
Sampled training rows	9,081, selecting every fourth row
Inner chronological split	7,265 fit / 1,816 development
Candidate configurations	5 squared-error penalties + 4 Huber configurations
Selected readout	Huber threshold 10°, penalty $\alpha = 100$
Final optimization	96 L-BFGS-B iterations
Bootstrap	100-frame blocks, 1,000 replicates, seed 20261021
MAE 95% interval	11.631° to 18.629°
RMSE 95% interval	18.241° to 35.675°
Paired MAE reduction interval	6.849° to 10.250° versus original RGB CNN

Appendix H Reference architecture counts

Canonical DAVE-2 reference. The local saved CNN has 1,595,511 parameters.

Layer	Parameters	Output
Input (YUV)	–	66×200×3
Normalisation	–	66×200×3
Conv1: 24 filters, 5×5, stride 2	1,824	31×98×24
Conv2: 36 filters, 5×5, stride 2	21,636	14×47×36
Conv3: 48 filters, 5×5, stride 2	43,248	5×22×48
Conv4: 64 filters, 3×3	27,712	3×20×64
Conv5: 64 filters, 3×3	36,928	1×18×64
Flatten	–	1,152
FC 100	115,300	100
FC 50	5,050	50
FC 10	510	10
Output (steering)	11	1
Total DAVE-2	252,219	
YOLO11s detector (10 classes)	9,416,670	100 layers, 21.4 GFLOPs
YOLO11s-seg lane network (1 class)	10,067,203	113 layers, 32.9 GFLOPs